A Literature Review of CNN-Based Computer Vision in Medical Imaging

Multi-Class Classification of Alzheimer's Disease from MRI Data

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Reviewed Paper:
El-Assy, A.M., Amer, H.M., Ibrahim, H.M., Mohamed, M.A. (2024).
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Abstract

Convolutional Neural Networks (CNNs) have become the dominant approach for automated analysis of medical images, particularly in neuroimaging-based diagnosis of neurodegenerative disorders such as Alzheimer's disease (AD). This review critically examines El-Assy *et al.* [1], published in *Scientific Reports* (2024), who propose a dual-branch CNN architecture for multi-class classification of AD stages from structural MRI. The review discusses the research problem, dataset, architecture, training strategy, evaluation protocol, and reported results, situates the work within the broader CNN-based medical imaging literature, and offers a critical assessment of its strengths, limitations, and directions for future research.

1 Introduction

Alzheimer's disease (AD) is a progressive neurodegenerative disorder and one of the leading causes of mortality among older adults worldwide. Its clinical progression spans a continuum from cognitively normal (CN) function through early and late mild cognitive impairment (EMCI, LMCI) to full dementia. Timely and accurate staging is critical for treatment planning, yet conventional diagnosis relies on manual interpretation of magnetic resonance imaging (MRI) by expert radiologists, a process that is time-consuming, subjective, and difficult to scale. Over the past decade, deep learning, and CNNs in particular, has emerged as the dominant paradigm for automating this task, motivated by the ability of convolutional architectures to learn hierarchical spatial features directly from raw imaging data without hand-crafted feature engineering [2].

Early work in this space largely relied on pretrained architectures such as AlexNet, VGG16, and ResNet variants, fine-tuned on MRI data [3,4], or on binary classification formulations that distinguish only AD from cognitively normal subjects. More recent research has shifted toward custom architectures and finer-grained multi-class formulations that better reflect the clinical reality of a disease continuum. This review focuses on one such contribution, El-Assy *et al.* [1], which proposes a purpose-built dual-branch CNN for 3-way, 4-way, and 5-way classification of AD stages using the ADNI dataset, and situates it against the wider body of CNN-based medical imaging research.

2 Related Work

The application of CNNs to AD diagnosis has followed a broadly similar trajectory to CNN-based medical imaging more generally: an initial reliance on transfer learning from ImageNet-pretrained networks, followed by domain-specific architectural innovation once the limitations of transfer learning in medical contexts became apparent. Ramzan *et al.* [3] employed ResNet-based residual networks on resting-state fMRI data for multi-class AD staging, while Fu'adah *et al.* [4] adapted AlexNet for MRI-based AD classification, reporting approximately 95% accuracy. Odusami *et al.* [5] extended this line of work with a hybrid pretrained-CNN model incorporating deep feature concatenation and gradient-weighted class activation mapping, while Shamrat *et al.* [6] proposed AlzheimerNet, a modified InceptionV3 model achieving 98.67% test accuracy on brain MRI.

A recurring limitation identified across this literature is the domain gap between natural images (on which these backbones are pretrained) and medical images, which can result in low transfer efficiency and a heightened risk of overfitting given the comparatively small size of most labelled medical imaging datasets [1]. A second recurring limitation is that a large proportion of prior studies restrict evaluation to binary classification (e.g., AD vs. CN), which does not reflect the graded, multi-stage nature of AD progression and is of limited clinical utility for early intervention. It is against this backdrop that El-Assy *et al.* [1] motivate a custom, non-pretrained CNN architecture explicitly designed for multi-class staging.

3 Overview of the Reviewed Study

3.1 Research Problem and Dataset

El-Assy *et al.* [1] address the problem of automated, fine-grained multi-class classification of AD stages, an area they argue is comparatively under-served relative to binary AD/CN classification. The study uses T1-weighted structural MRI scans from the Alzheimer's Disease Neuroimaging Initiative (ADNI), comprising 1,296 3D scans at 1.5 mm isotropic voxel resolution, labelled across five diagnostic categories: CN, EMCI, MCI, LMCI, and AD. To address class imbalance, the authors apply ADASYN (Adaptive Synthetic Sampling) [8], an oversampling technique that generates synthetic minority-class samples in proportion to classification difficulty, expanding the effective dataset to approximately 3,000 images. Standard geometric and photometric augmentation (cropping, scaling, flipping, brightness/contrast adjustment) is also applied, and the data is partitioned 90%/10% for training and testing, with cross-validation used within the training split for hyperparameter selection.

3.2 Proposed Architecture

The central methodological contribution is a dual-branch CNN in which two independently designed sub-networks are concatenated at the classification layer, rather than the ensembling of pretrained backbones seen in prior work [5, 7]. The first branch (CNN-1) uses successive convolutional blocks with 3×3 filters (16, then 64, then 256 filters), interleaved with 2×2 max-pooling. The second branch (CNN-2) mirrors this depth but uses larger 5×5 filters (32, then 128, then 512 filters). Both branches apply 20% dropout and batch normalization before a 128-unit fully connected layer, and their outputs are concatenated and passed through a final classification layer. Figure 1 illustrates the overall flow of the proposed dual-branch architecture.

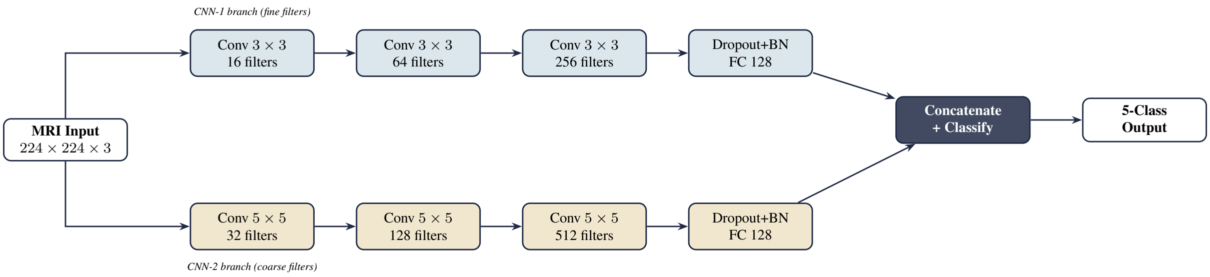

Figure 1: Overview of the dual-branch CNN architecture proposed by El-Assy *et al.* [1]. Two parallel CNN branches with different filter sizes independently process the MRI input; their learned features are concatenated at the classification layer.

The authors report that the 3×3-filter branch reaches comparable accuracy to the 5×5-filter branch using roughly half the number of filters, and that the fused, dual-branch model consistently outperforms either branch in isolation, which they attribute to the two branches learning complementary spatial representations.

3.3 Training Strategy and Evaluation

The model is implemented in Keras with a TensorFlow backend and trained on a single Tesla K80 GPU via Google Colaboratory Pro. Overfitting is mitigated through dropout, batch normalization, and data augmentation, while class imbalance is addressed through ADASYN oversampling prior to training. Model selection uses cross-validation within the training set, with a held-out test set reserved exclusively for final evaluation. Performance is assessed using a comprehensive metric suite comprising accuracy, precision, recall, F1-score, balanced accuracy, and the Matthews Correlation Coefficient (MCC), together with per-class ROC/AUC analysis and confusion matrices. Statistical significance of the improvement over baseline methods is verified using the Wilcoxon signed-rank test, and Grad-CAM [9] is used to visualize class-discriminative regions for qualitative interpretability.

3.4 Results

The fused dual-branch model achieves 99.43% accuracy on the 3-way classification task (AD/MCI/CN), 99.57% on the 4-way task, and 99.13% on the most challenging 5-way task (AD/CN/LMCI/EMCI/MCI). Ablation results show that fusion substantially improves upon either branch alone, with 5-way accuracy rising from approximately 95% for a single branch to 99.13% for the combined model, as summarized in Figure 2.

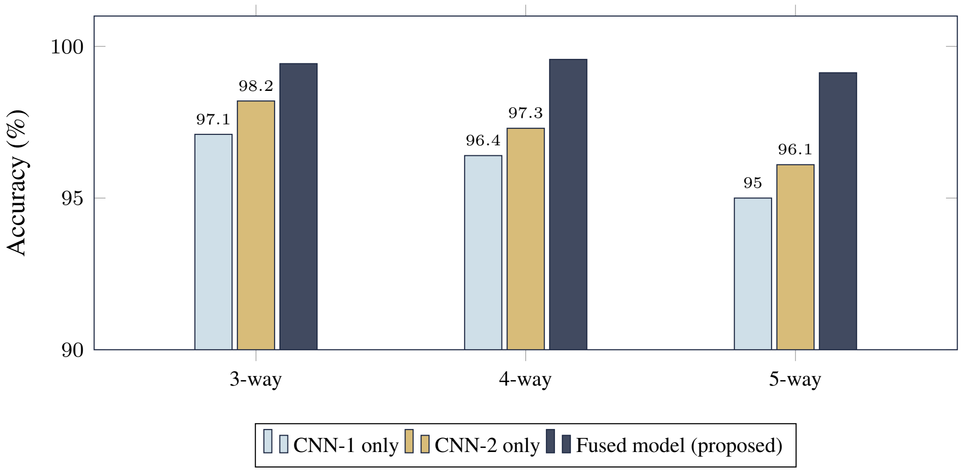

Figure 2: Reported classification accuracy of the individual CNN branches versus the fused dual-branch model across the 3-way, 4-way, and 5-way AD staging tasks [1]. The fused model consistently outperforms either branch alone.

Per-class AUC values are near-ideal across all five categories, and the confusion matrices indicate that the small number of misclassifications occur predominantly between clinically adjacent stages (e.g., CN misclassified as EMCI), which the authors argue represents a clinically preferable failure mode to missing a diseased case outright. The Wilcoxon test confirms that these improvements over four prior comparison methods are statistically significant ($p < 0.05$).

4 Critical Analysis

The reviewed study makes a genuine methodological contribution: rather than relying on transfer learning from non-medical pretrained networks, it demonstrates that two comparatively lightweight, purpose-built CNN branches with different receptive field sizes can be fused to outperform either branch individually, and the ablation study substantiates this claim rather than merely asserting it. The evaluation protocol is also more rigorous than much of the surrounding literature, combining imbalance-aware metrics (MCC, balanced accuracy), statistical significance testing, and Grad-CAM-based interpretability rather than relying on accuracy alone.

At the same time, the reported accuracies of 99.1–99.6% are notably higher than the 85–95% range typically reported for well-validated five-class AD-staging models elsewhere in the literature, which warrants scrutiny. A plausible source of this gap is the application of ADASYN oversampling prior to the train/test split (or prior to cross-validation), which risks introducing near-duplicate synthetic samples across partitions and inflating apparent performance; the methodology as described does not clearly rule this out. The dataset is also modest in absolute terms (1,296 original scans from a single, curated, multi-site research cohort), and single-dataset evaluation cannot establish whether the network is learning genuine anatomical biomarkers of disease progression or scanner- and site-specific acquisition artefacts that happen to correlate with diagnostic labels. Furthermore, the model relies exclusively on imaging data and does not incorporate the clinical, cognitive, or biomarker information that radiologists typically use alongside MRI in practice. These concerns do not diminish the architectural contribution of the paper, but they do suggest that the reported accuracy figures should be treated cautiously until replicated on an independent, externally sourced cohort with a leakage-safe resampling protocol.

5 Conclusion and Future Directions

El-Assy *et al.* [1] present a dual-branch CNN architecture that achieves strong reported performance on multi-class AD staging from MRI without relying on pretrained backbones, and pair this with an evaluation methodology that is more comprehensive than much of the comparable literature. The authors identify three directions for future work: validating the approach on larger and more diverse datasets to establish generalizability beyond ADNI, assessing feasibility for real clinical deployment rather than research benchmarking alone, and integrating additional modalities such as PET imaging and clinical or cognitive scores alongside MRI to further improve diagnostic accuracy. Addressing the leakage risk associated with the oversampling protocol, and validating results on an independent, multi-site cohort, would substantially strengthen confidence in the approach's clinical relevance.

References

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